

CS & IT ENGINEERING



Computer Network - 2

Transport & Application Layer

Lecture No. - 04

By - Abhishek Sir





Recap of Previous Lecture



Topic

TCP Options

Topic

TCP Window Scale

Topic

TCP Timers



Topics to be Covered



Topic

TCP Timers

Topic

Application Layer

Topic

DNS

[MCQ] [GATE-2015, 2-Mark]

#Q. Assume that the bandwidth for a TCP connection is 1048560 bits/sec. Let α be the value of RTT in milliseconds [(rounded off to the nearest integer)] after which the TCP window scale option is needed. Let β be the maximum possible window size with window scale option. Then the values of α and β are.

- ☒ A 63 milliseconds, 65535×2^{14}
- ☒ B 63 milliseconds, 65535×2^{16}
- ☒ C 500 milliseconds, 65535×2^{14}
- ☒ D 500 milliseconds, 65535×2^{16}

Max^m RTT (without window scale)

$$\alpha = \frac{(2^{16} - 1) \text{ bytes}}{\text{Bandwidth}} = \frac{(2^{16} - 1) * 2^3 \text{ bits}}{1048560 \text{ bits/sec}}$$

$$\alpha = 0.5 \text{ sec} = 500 \text{ ms}$$

Max^m RWND (with window scale)

$$\beta = (2^{16} - 1) * 2^{14} \text{ bytes}$$

Ans: C

[MCQ] [GATE-IT-2007, 1-Mark]



#Q. Consider the following statements about the timeout value used in TCP.

- i. The timeout value is set to the RTT (Round Trip Time) measured /during TCP connection establishment for the entire duration of the connection. FALSE
- ii. Appropriate RTT estimation algorithm is used to set the timeout value of a TCP connection. TRUE
- iii. Timeout value is set to twice the propagation delay from the sender to the receiver. FALSE

Which of the following choices hold?

- ☐ A (i) is false, but (ii) and (iii) are true
- ☒ B (i) and (iii) are false, but (ii) is true
- ☐ C (i) and (ii) are false, but (iii) is true
- ☐ D (i), (ii) and (iii) are false

Ans: B

[MCQ] [GATE-2015, Set-1, 1-Mark]



#Q. Suppose two hosts use a TCP connection to transfer a large file. Which of the following statements is/are False with respect to the TCP connection?

- I. If the sequence number of a segment is m , then the sequence number of the subsequent segment is always $(m+1)$. FALSE
- II. If the estimated round trip time at any given point of time is t sec, the value of the retransmission timeout is always set to greater than or equal to t sec. TRUE
- III. The size of the advertised window never changes during the course of the TCP connection. FALSE, [RWND updated dynamically]
- IV. The number of unacknowledged bytes at the sender is always less than or equal to the advertised window TRUE

A III only

☒ B I and III only

C I and IV only

☐ D II and IV only

Ans: B



Topic : Keepalive Timer



→ Prevents long idle connections between two TCPs

→ Each time TCP hears from other TCP, it reset the timer

→ if TCP does not hear from other TCP till keepalive timer time-out, the TCP sends a special segment called "probe"

→ This prevents from indefinite wait



Topic : TCP



TCP Sender

TCP Receiver

TCP Segment
[Payload : Data]

[ACK]

TCP connection
state lost;

[wait for
next TCP
data segment]
[Indefinite
Wait]

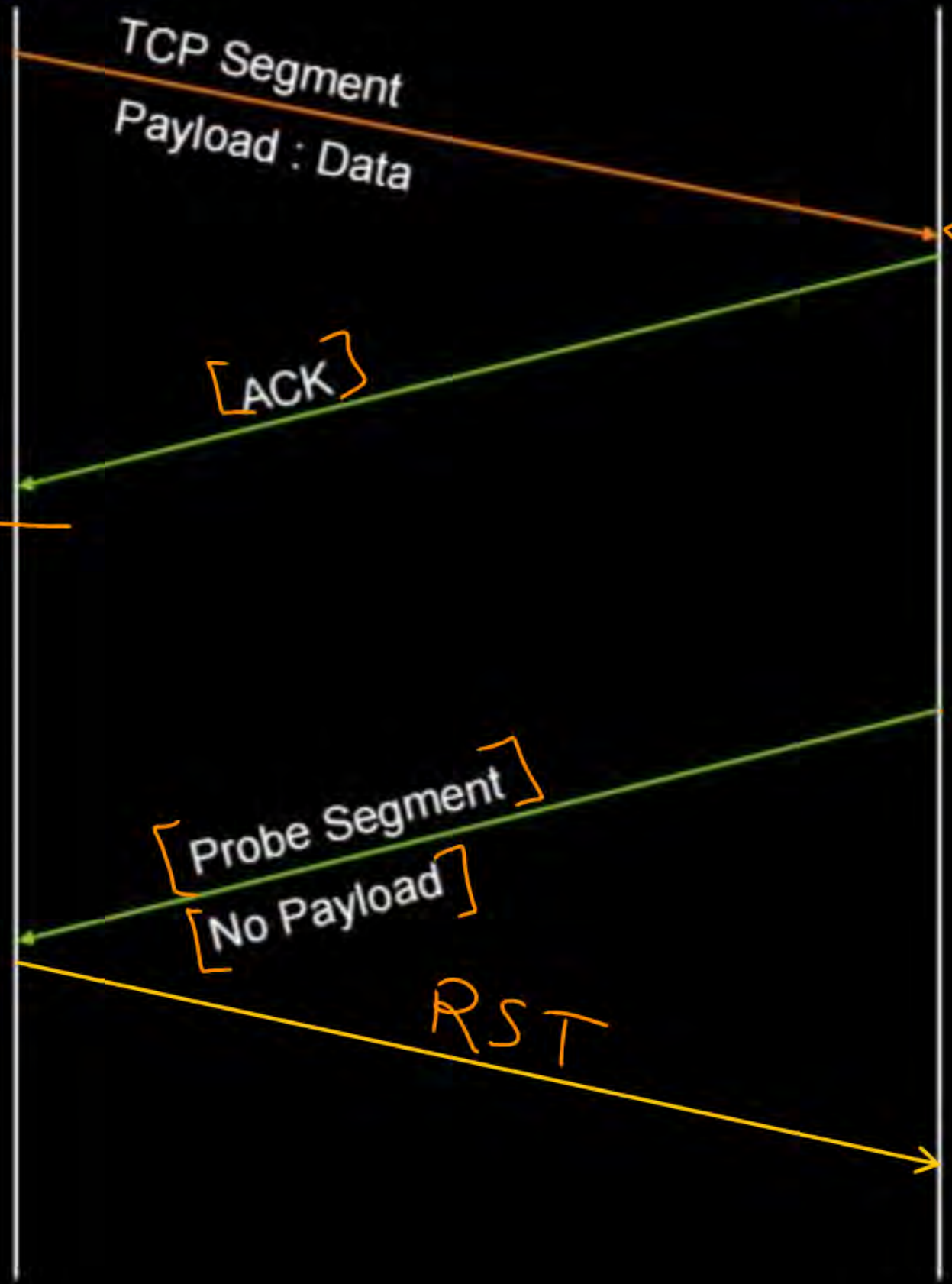


Topic : TCP



TCP Sender

TCP Receiver



[TCP Conn. lost]

[K.T. Reset]

---T.O. [KT]

[Probe Segment]
[No Payload]

RST

[MSQ] [GATE-2021, 2-Marks] IIT-B, H.W.

#Q. A TCP server application is programmed to listen on port number P on host S. A TCP client is connected to the TCP server over the network. Consider that while the TCP connection was active, the server machine S crashed and rebooted. Assume that the client does not use the TCP keepalive timer. Which of the following behaviors is/are possible?

- ☐ A If the client was waiting to receive a packet, it may wait indefinitely
- ☐ B The TCP server application on S can listen on P after reboot
- ☐ C If the client sends a packet after the server reboot, it will receive a RST segment
- ☐ D If the client sends a packet after the server reboot, it will receive a FIN segment



Topic : Time-waited Timer



- Used during connection termination
- More than twice the MSL $\boxed{\geq 2MSL}$
- When TCP closes a connection, it is not really closed, until this timer time-out
- Allow FIN segment to be received



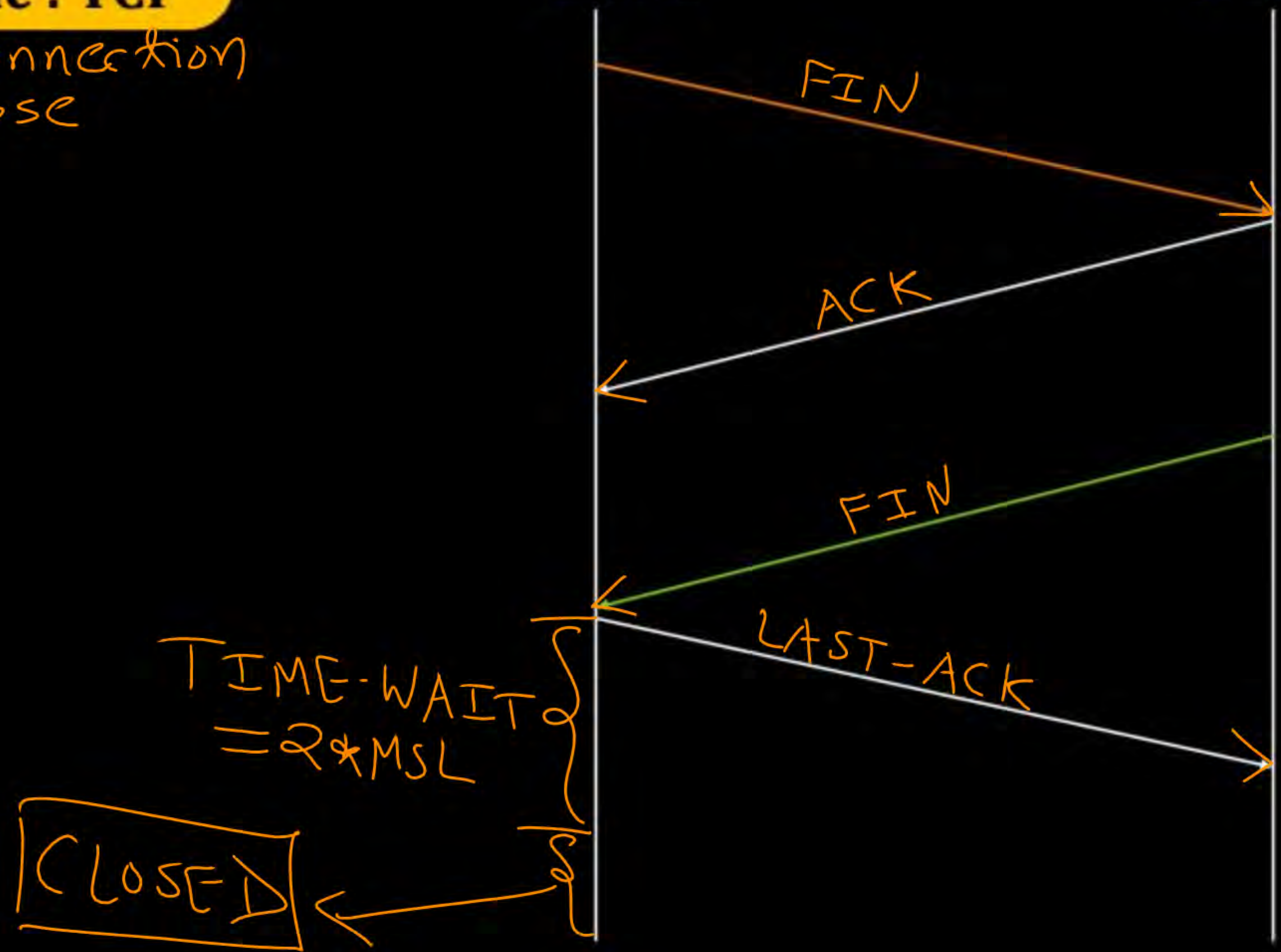
Topic : TCP



⇒ Connection close

TCP Client

TCP Server





Topic : Delayed Timer ✓



→ Method to optimize network traffic ✓
[by delaying acknowledgments for received data segments]

→ Instead of immediately sending an individual ACK for every segment
[allowing the ACK to piggyback on the next outgoing segment]

→ TCP waits for a short period using delayed timer
[waiting for another segment will arrive, till timeout]



Topic : TCP

TCP Client

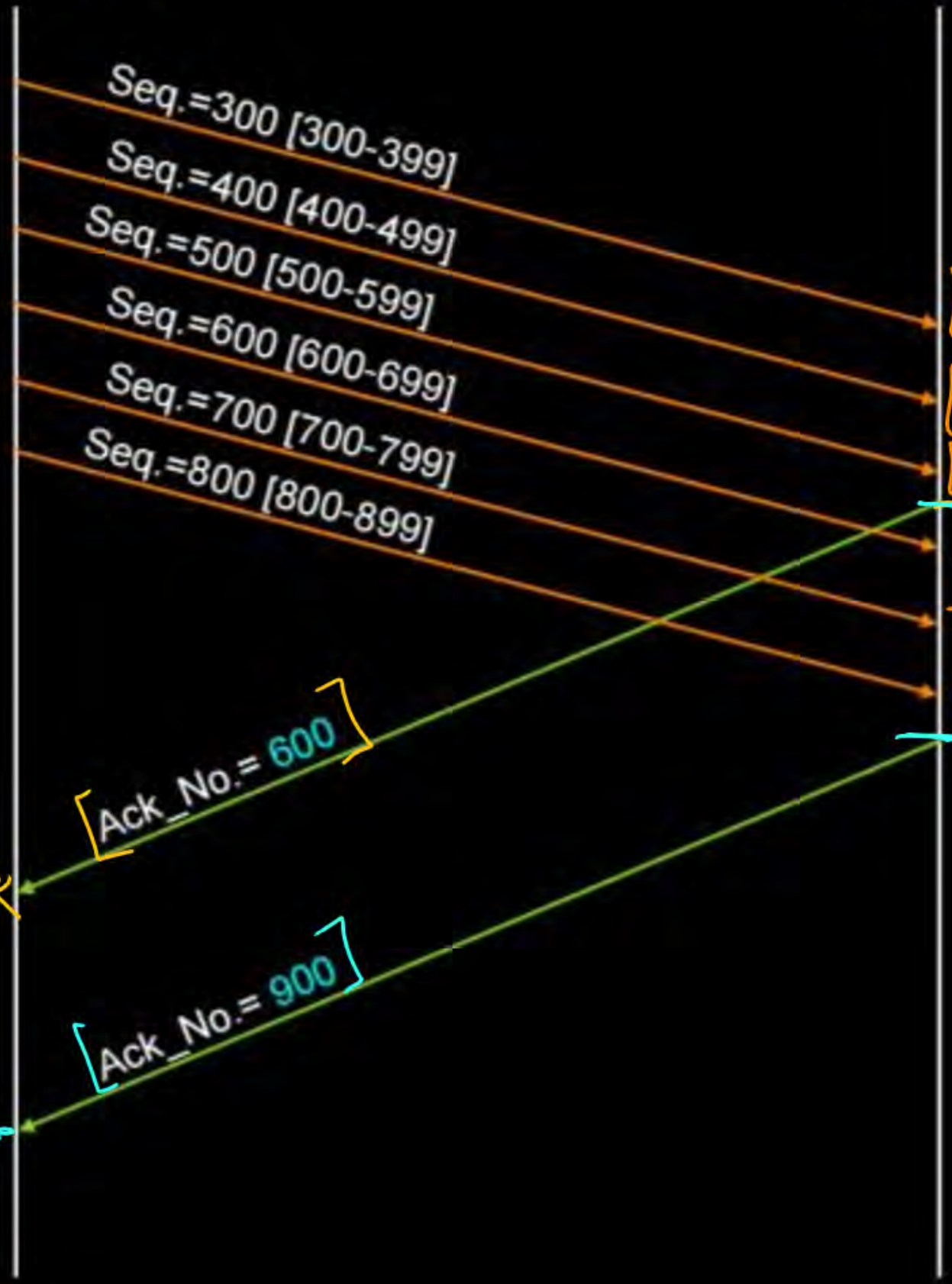
TCP Server



Client_ISN = 299

MSS = 100 bytes

Server_ISN = 1599



Till byte 599, combine all Acks
Till byte 899 Acks

NBE =
~~300~~ ~~700~~
~~400~~ ~~800~~
~~500~~ 900
~~600~~

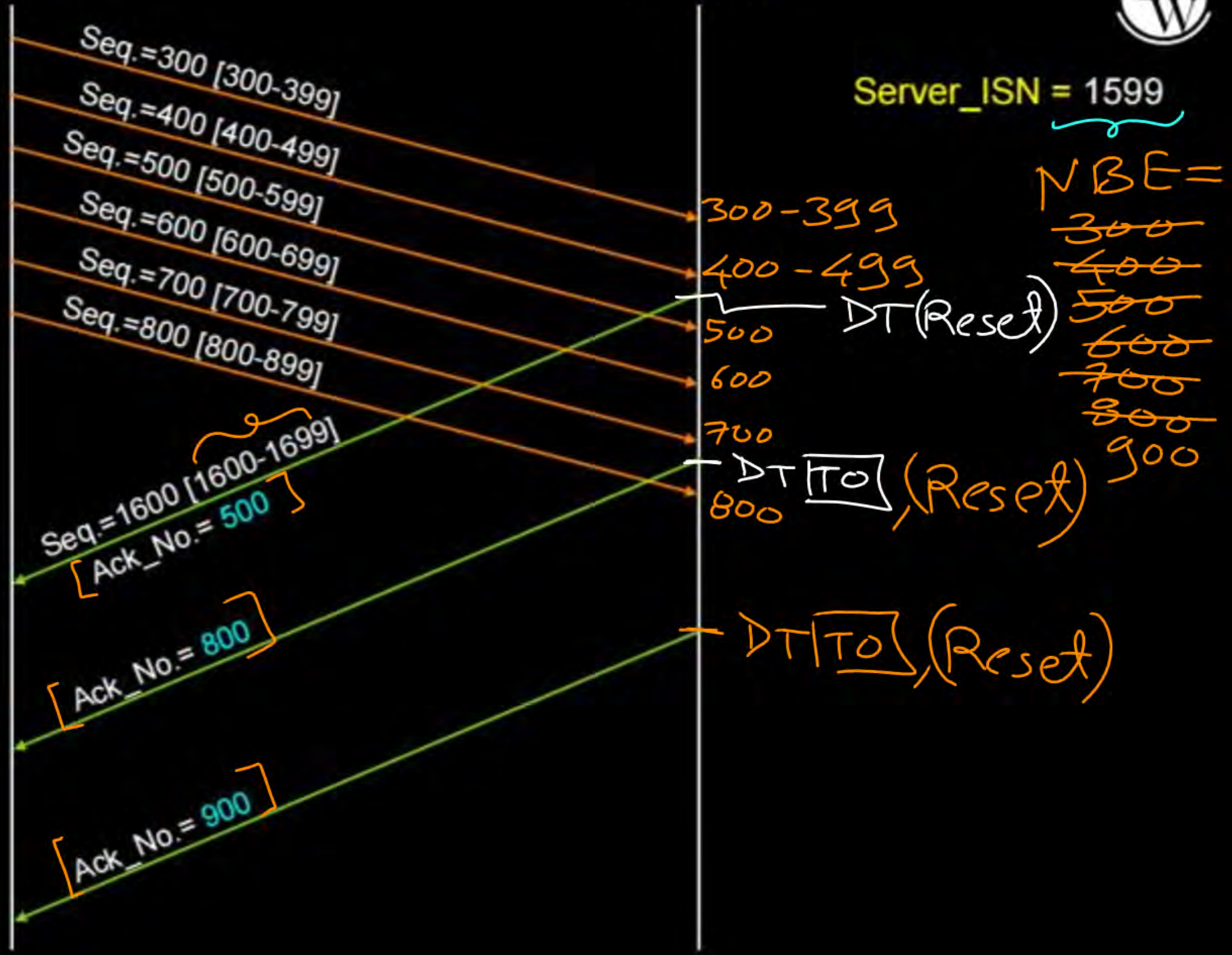


Topic : TCP

Client_ISN = 299
MSS = 100 bytes

TCP Client

TCP Server





Topic : Silly Window Syndrome (SWS)

→ When TCP receiver advertises smaller value in 'Window Size'

→ And TCP sender sends smaller size TCP segment
[as per advertised smaller value in 'Window Size']

→ Lead more overhead in the network ⇒ Degrade TCP performance



Topic : TCP

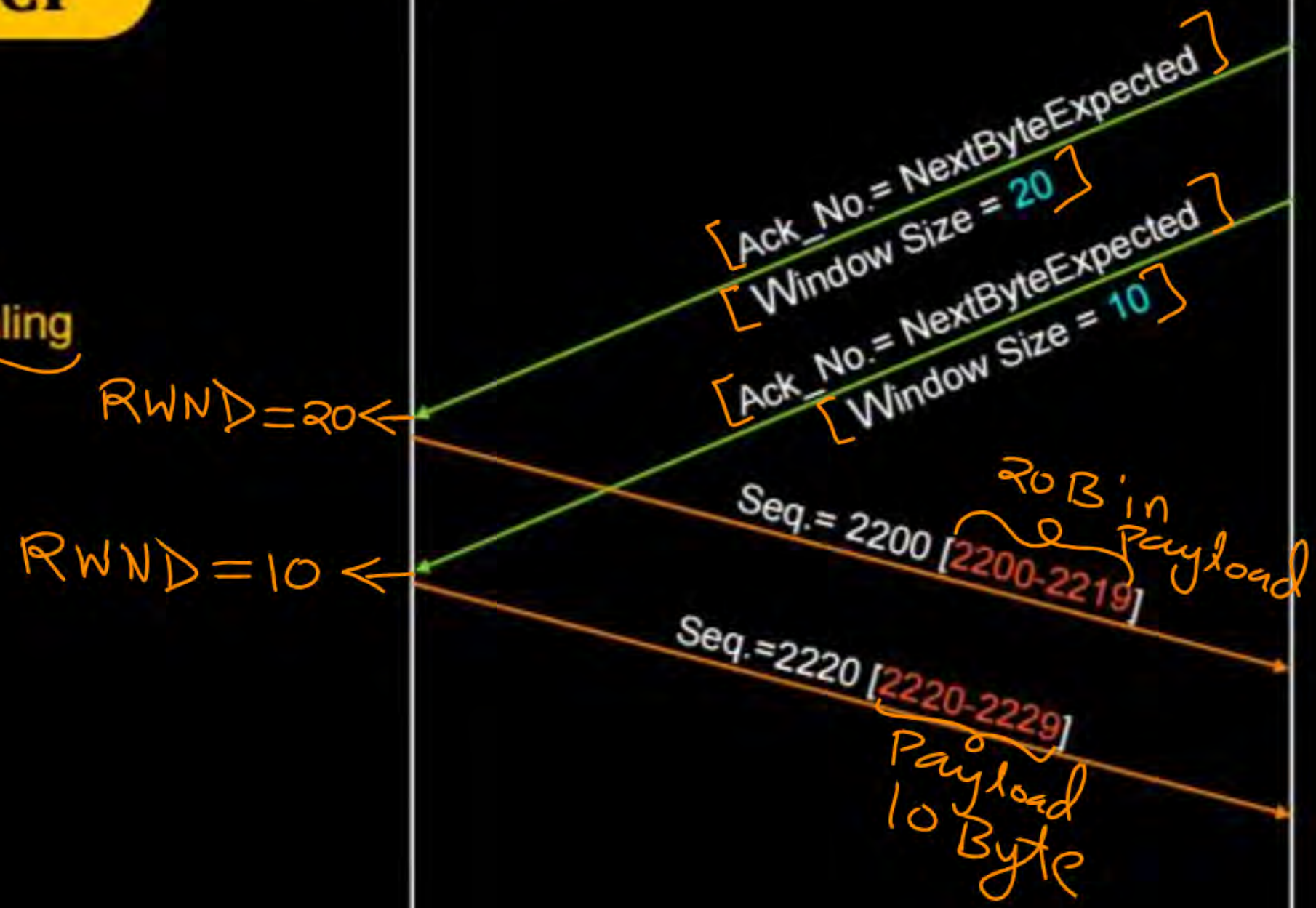
TCP Client

TCP Server



MSS = 500 bytes

Without window scaling





Topic : TCP

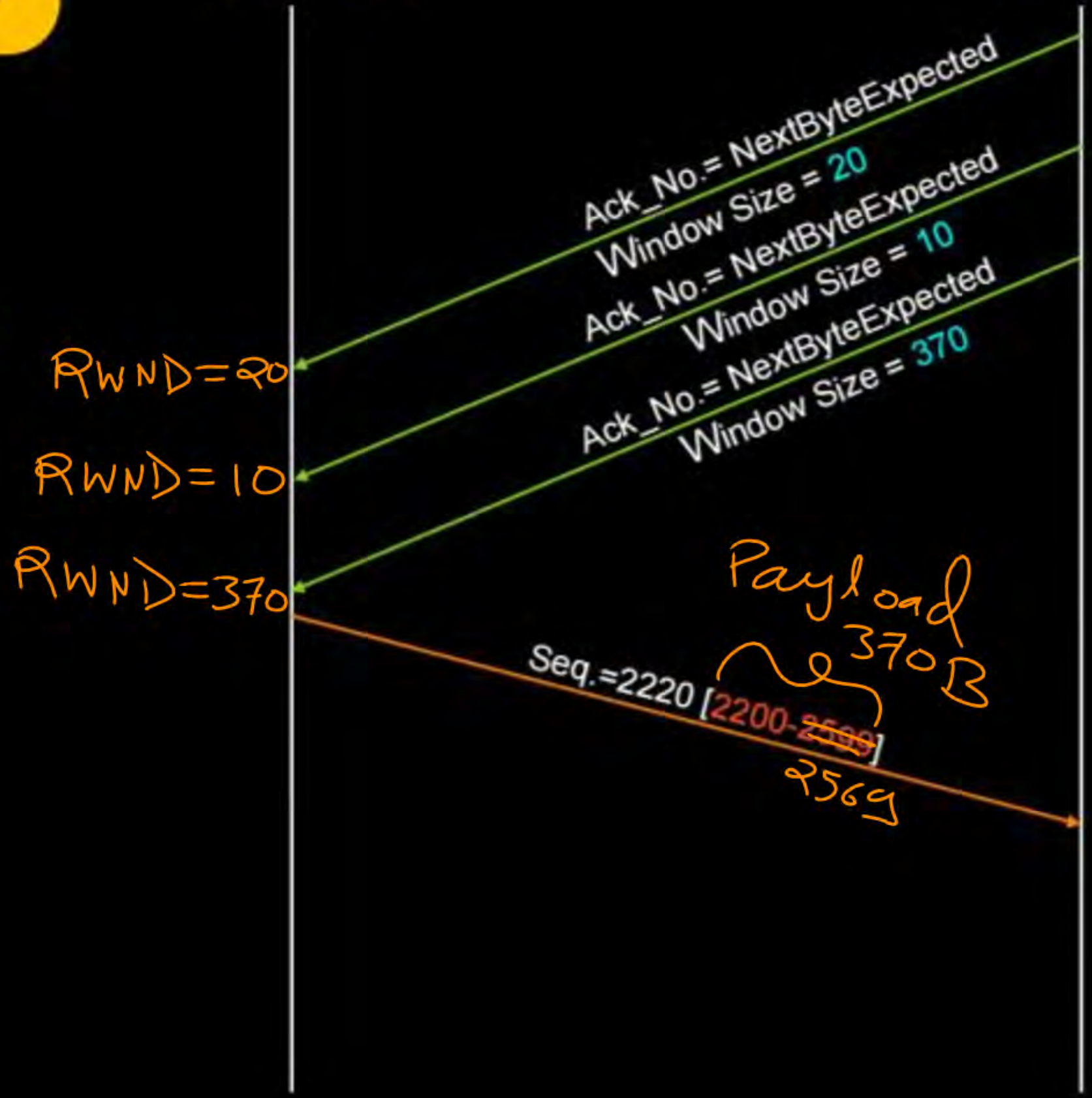
TCP Client

TCP Server



MSS = 500 bytes

Without window scaling





Topic : TCP

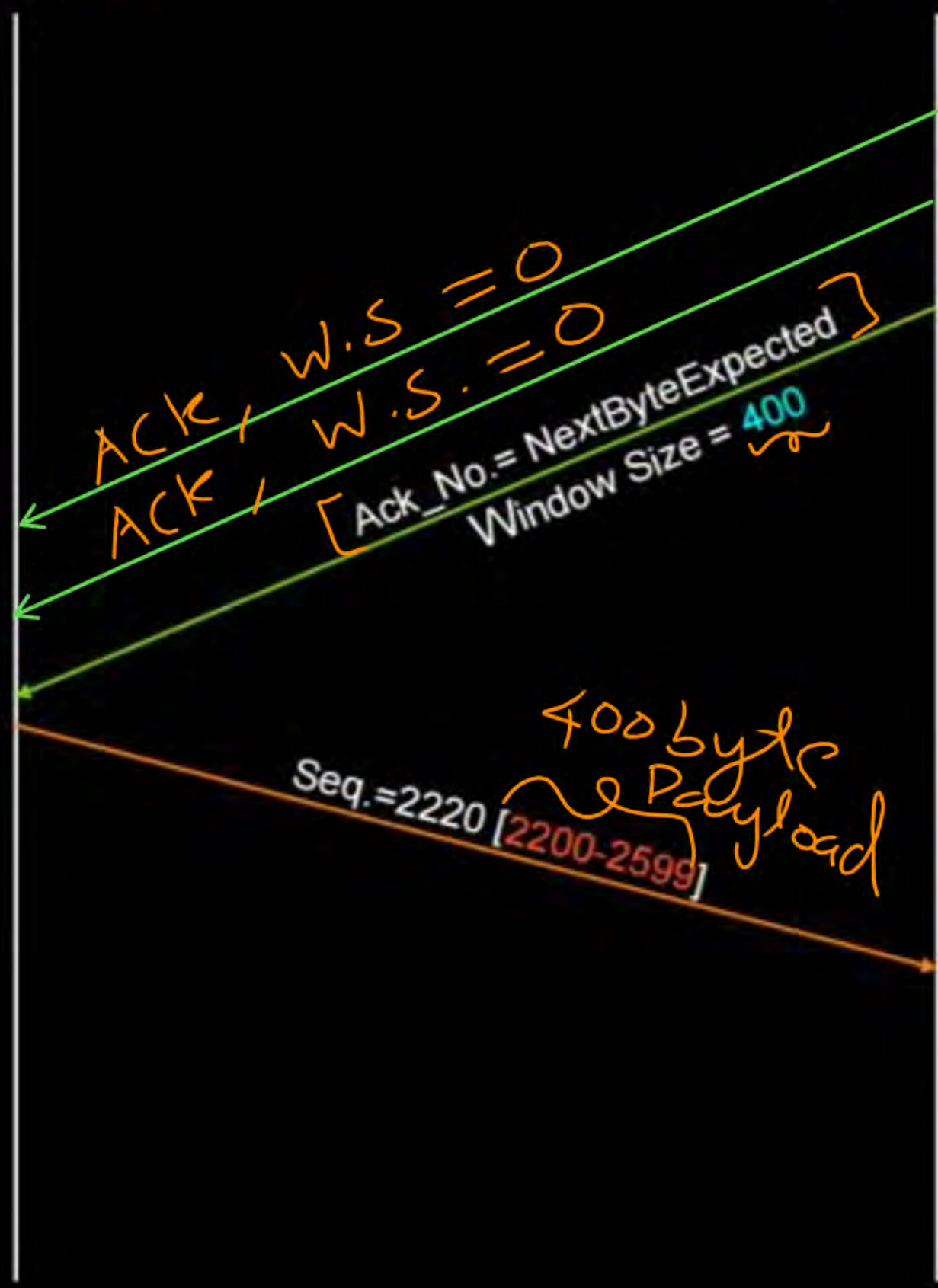
TCP Client

TCP Server



MSS = 500 bytes

Without window scaling



[MCQ] [ISRO-2007, 1-Marks]

#Q. Silly Window Syndrome is related to

- ☐ A Error during transmission
- ☐ B File transfer protocol
- ☒ C Degrade in TCP performance
- ☐ D Interface problem

Ans: C



Topic : Urgent Pointer



→ Urgent Pointer (16-bits)

→ Valid only if URG (urgent) flag is set (on)

→ Defines the number to be added to the sequence number to determine the last urgent byte in the payload



Topic : Push Flag



→ Do not buffer the data in 'Send Buffer'
[Source process inform TCP, data should be sent immediately]

→ Push it immediately into the channel,
[bypassing the buffer and reducing latency]

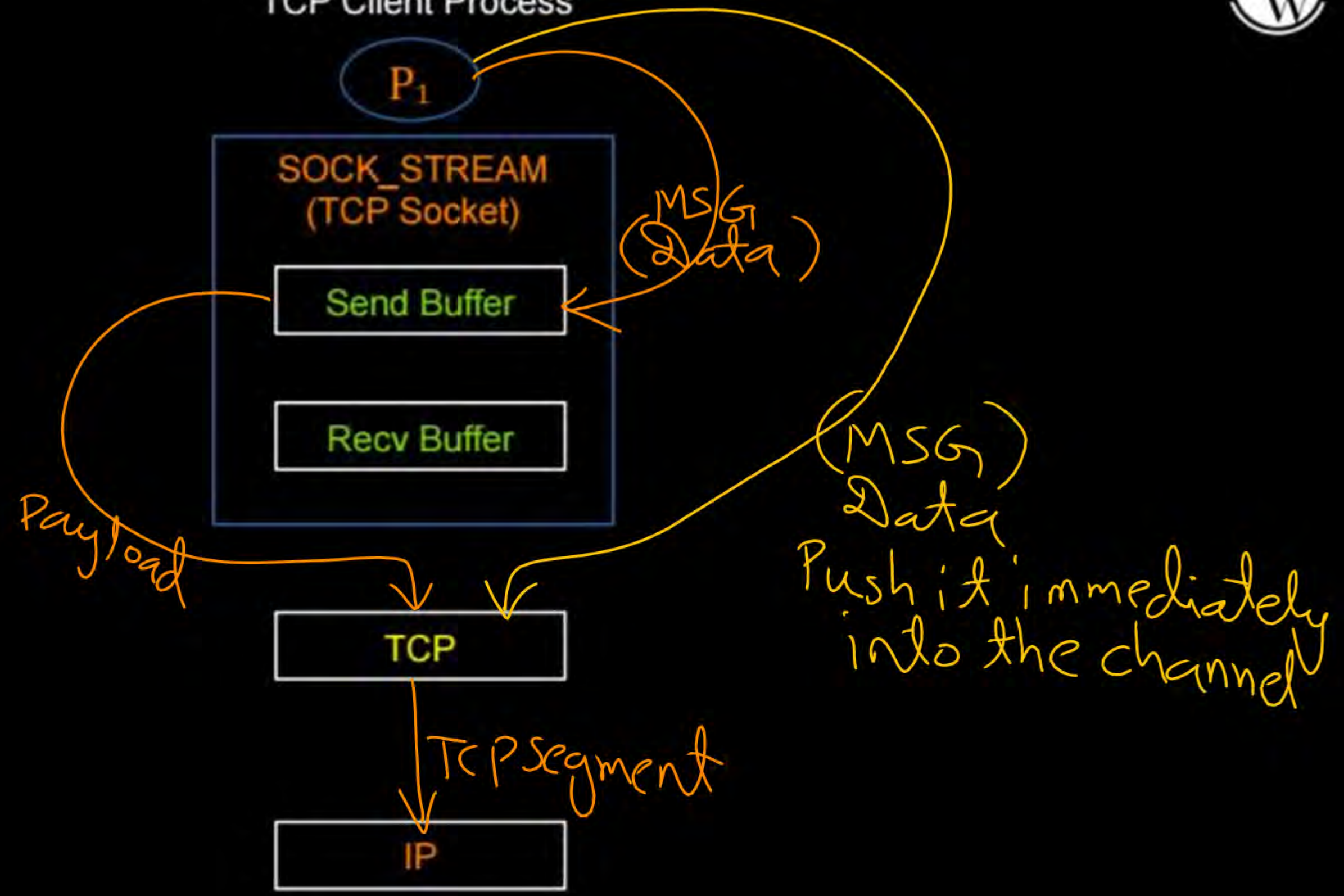
→ Set Push Flag ON, in the sending TCP segment



Topic : TCP



TCP Client Process





Topic : Push Flag



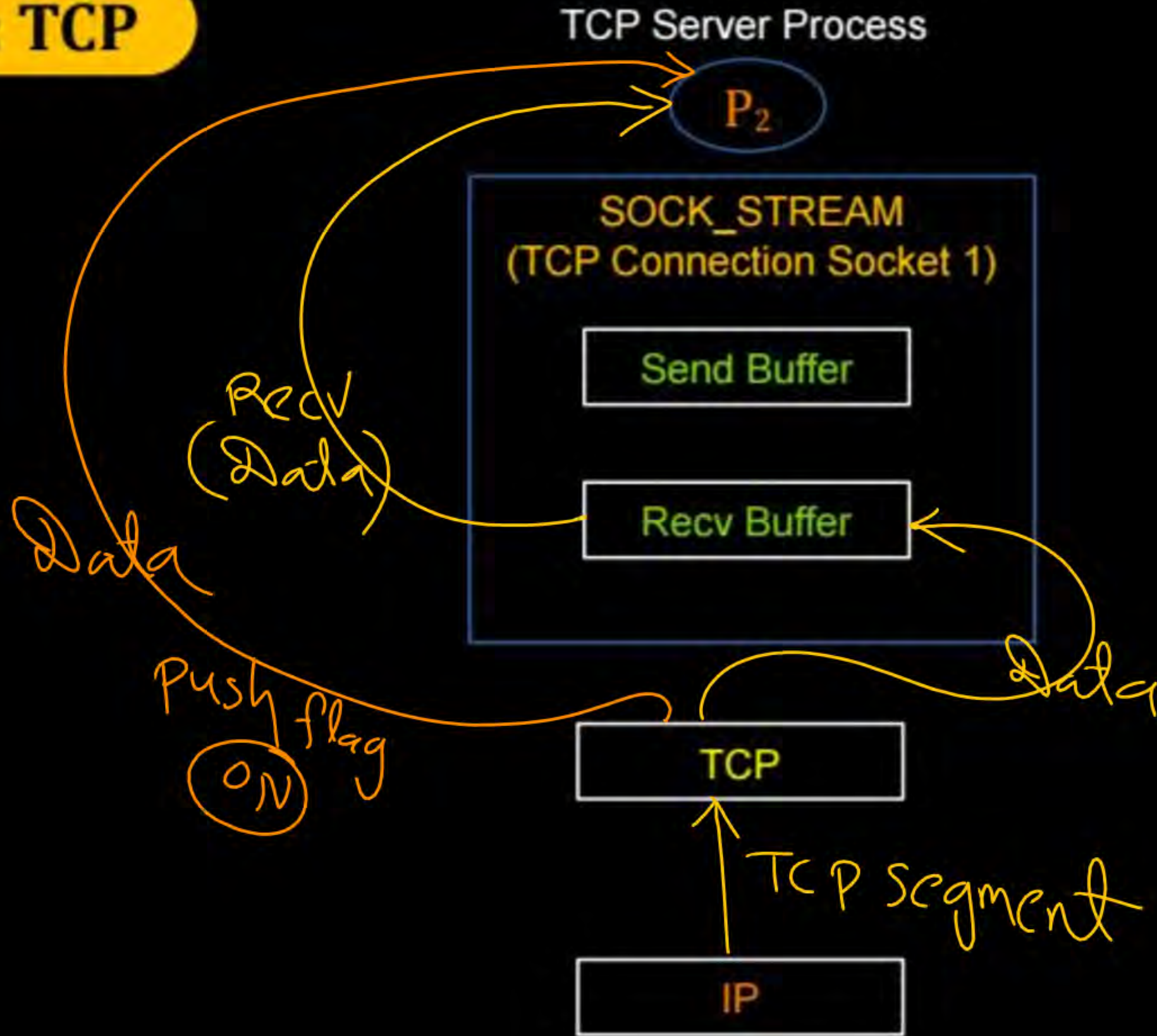
→ if Push Flag is ON, in the received TCP segment

→ Do not buffer the data in 'Recv Buffer'
[signals the receiver to immediately forward the received data to the application]

→ Push it immediately to the application
[bypassing the buffer and reducing latency]



Topic : TCP





Topic : Syllabus



Transport layer : flow control and congestion control, UDP, TCP, sockets;

TCP/AIMD
↑
V.V.Imp

V.V.Imp.
↑
✓

Application layer protocols : DNS, SMTP, HTTP, FTP, Email.

Imp.
V.Imp



Topic : Application Layer



- Network application program (network process)
[e.g. Browser]
- Process : Program in execution



Topic : Port Number



- Port Number (16-bit) : Identifier $\Rightarrow$ Local [within a host unique]
- Used to identify a **network process** in a machine
[Process involved in **Network Communication**]



Topic : Two Process Communication



→ Two network processes, belongs to different hosts

Communicate over network, by exchanging "messages" (Data)

OSI Model

TCP/IP

App. Layer PDU



Topic : Two Process Communication



Communication Credentials :

→ Source Port No. (16-Bit)

→ Source IPv4 Address (32-Bit)

→ Destination IPv4 Address (32-Bit)

→ Destination Port No. (16-Bit)



Topic : Two Process Communication



Two paradigm :

1. **Client - Server**

[e.g. Web browsing, Email]

2. **Peer to peer** ✓

[e.g. BitTorrent, VoIP]



Topic : Client - Server Paradigm



Server :

- Always on host
- Permanent IP Address
- Mostly in data centers



Topic : Client - Server Paradigm



Client :

- Communicate (contact) with server
- May have dynamic IP address
- Clients do not communicate directly with each other in client-server Model



Topic : Client - Server Paradigm



Example :-

1. Web browsing

Client : Web browser *web client*

Server : Web Server

2. Email

Client : User Agent

Server : Mail Server



Topic : Peer - Peer Architecture



- No always on server
- Arbitrary end systems directly communicate
- Complex management (Peers can change IP address)
- e.g. P2P file sharing (BitTorrent), Voice over IP (VoIP)

→ Internet based voice/
Video
Call

[MCQ] [GATE-2010, 1-Marks] IIT-G

#Q. Which one of the following is not a client server application ?

- ☒ A Internet chat
 - ☒ B Web browsing
 - ☒ C E-mail
 - ☐ D Ping
- } client-server

Ans: D

Ans : (D) Ping

Ping is utility, to check connectivity between either client - client or client - server.



Topic : Protocol



→ Set of Rules

→ Define, how data is exchanged between network processes



Topic : Application Layer



Application Layer Protocols : [client-server Model]

- DNS : Domain Name System
- HTTP : Hyper Text Transfer Protocol
- FTP : File Transfer Protocol
- SMTP : Simple Mail Transfer Protocol



Topic : Stateless vs Stateful Protocol

⇒ Cookies



→ Server maintains no information about past client requests
[Every request from client will be treated as new request by server]

→ Protocols that maintain "state" are complex
[Past history (state) must be maintained]

Transport Layer = UDP (stateless), TCP (stateful)

Application Layer: stateless = DNS, HTTP, SMTP

stateful = HTTPS, FTP, IMAP (POP) SMTP



Topic : DNS



- Domain Name System (DNS)
- 'Stateless' Application Layer Protocol
- Client - Server Model



Topic : DNS



- Used to translate 'Host Name' into 'IP Address' (vice versa)
→ (Web/Email) server name
- Hosts (DNS clients) and DNS server communicate to resolve names
[Name / Address translation]



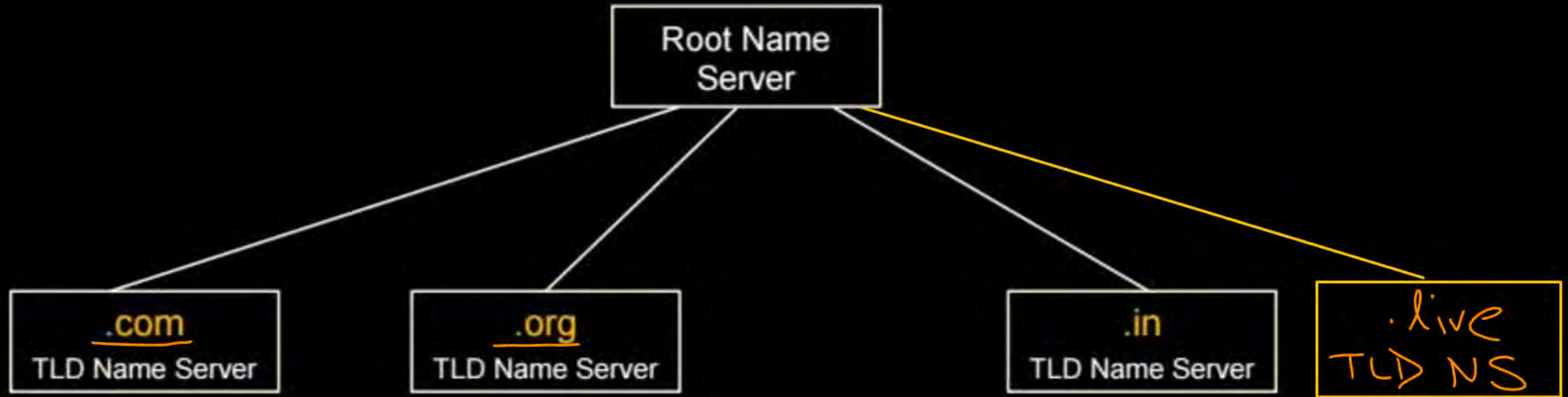
Topic : DNS Hierarchy



- Distributed Hierarchical Database : [Global Hierarchy] ✓
[Implemented in hierarchy of many name servers]



Topic : DNS Hierarchy





Topic : DNS Hierarchy



→ DNS name servers :

1. Root Name Servers ↓
2. Top-Level Domain (TLD) Name Servers ↓
3. 3. Authoritative Name Servers



Topic : Root Name Servers



→ Point of contact for any DNS query

→ ICANN manages root DNS domain

→ 13 logical root name servers world wide
[each server replicated many times]



Topic : TLD Name Servers

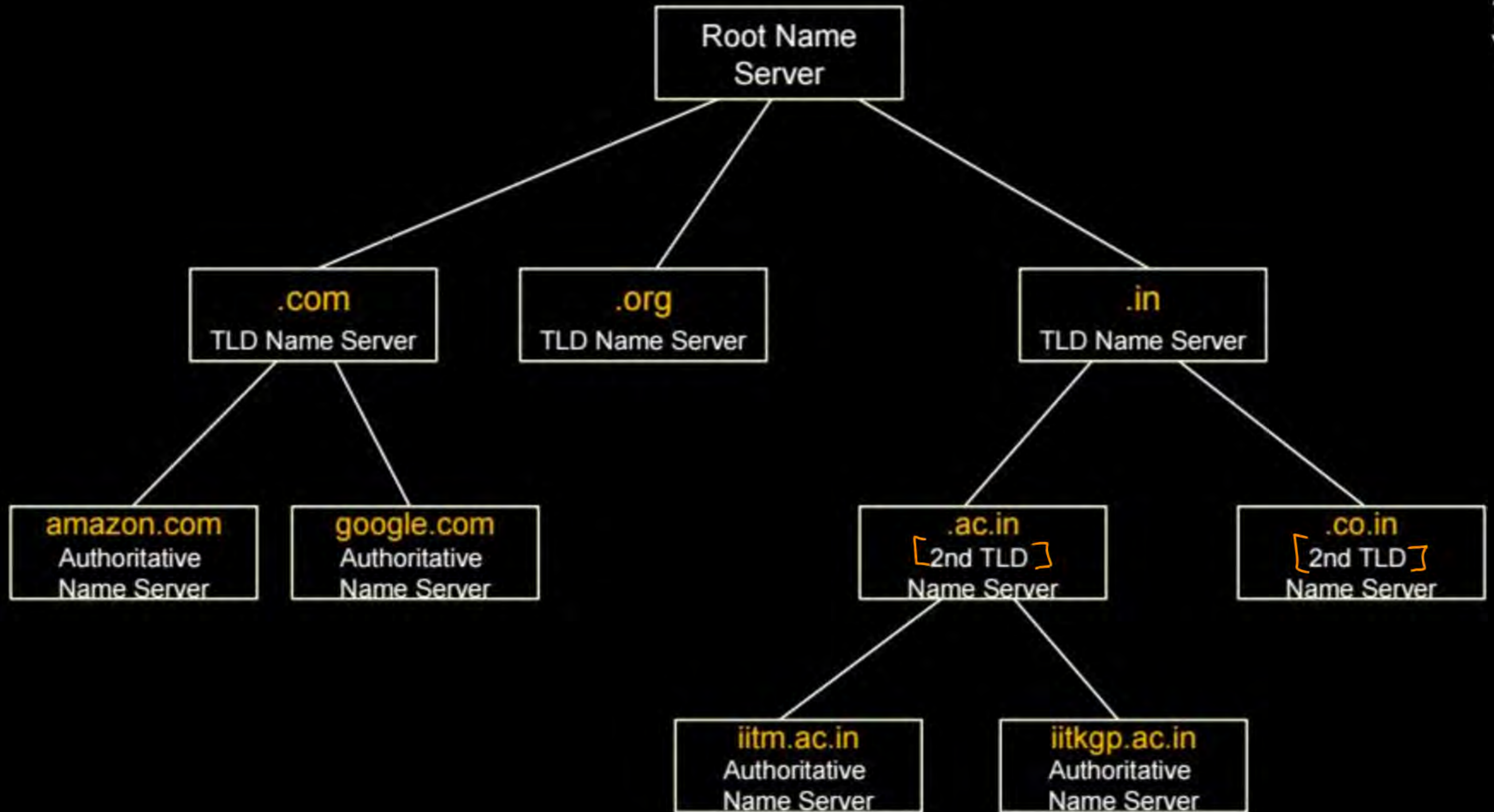


→ Responsible for all top level domains like ".com", ".org" etc
[Country domains e.g. ".in", ".uk", etc]

⇒ Some DNS hierarchy are 3-tier

⇒ Some are 4-tier

* Intermediate (2nd) TLD





2 mins Summary



Topic

TCP Timers ✓

Topic

Application Layer ✓

Topic

DNS →



THANK - YOU

